

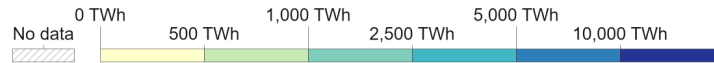
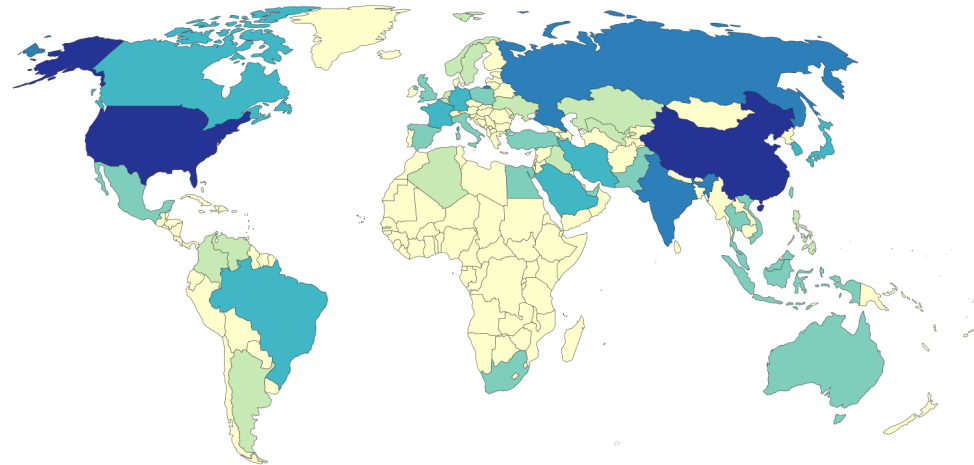
Energy Crisis

-Biofuel & Nanotechnology

Webb Wu

Primary energy consumption, 2021

Primary energy¹ consumption is measured in terawatt-hours (TWh).



Source: BP Statistical Review of World Energy; and EIA

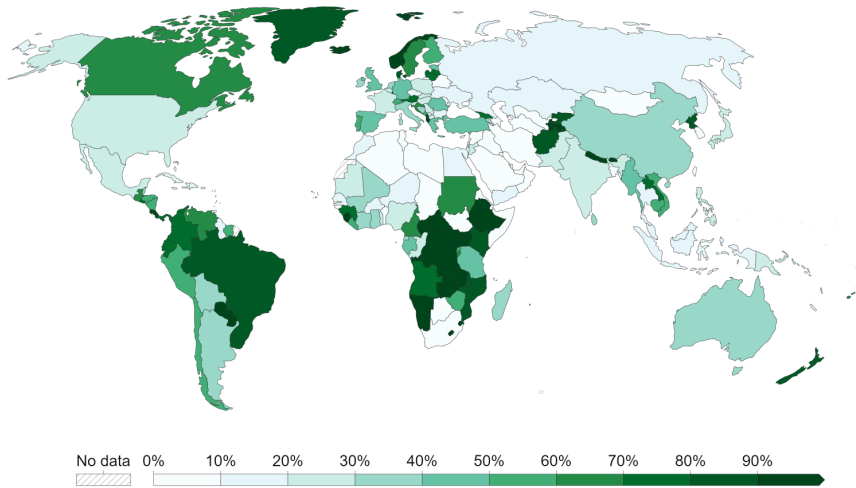
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Note: Data includes only commercially-traded fuels (coal, oil, gas), nuclear and modern renewables. It does not include traditional biomass.

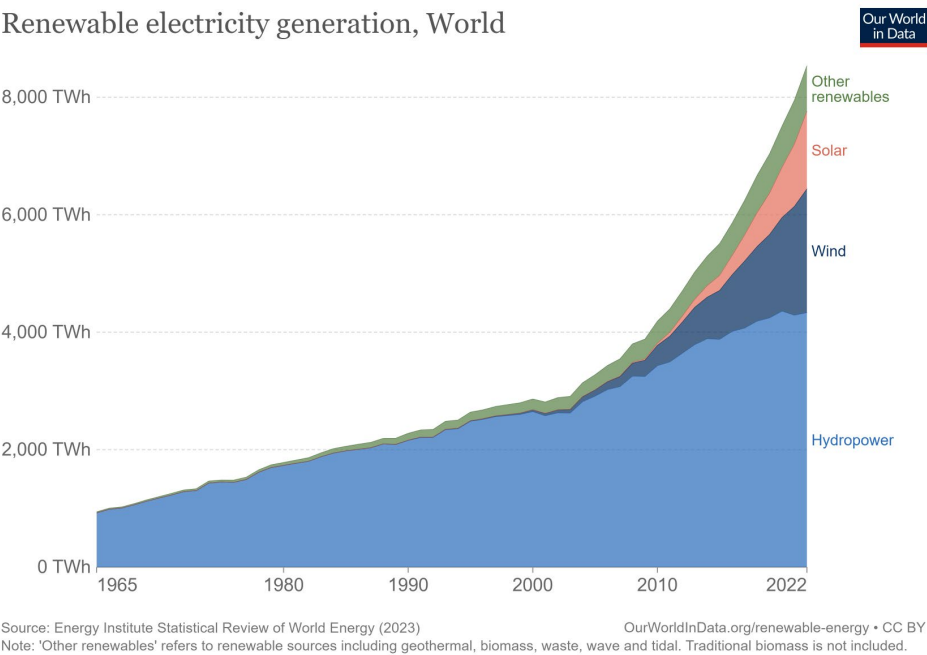
1. Primary energy: Primary energy is the energy as it is available as resources – such as the fuels that are burnt in power plants – before it has been transformed. This relates to the coal before it has been burned, the uranium, or the barrels of oil. Primary energy includes energy that is needed by the end user, plus inefficiencies and energy that is lost when raw resources are transformed into a useable form. You can read more on the different ways of measuring energy in our article.

Share of electricity production from renewables, 2022

Renewables include electricity production from hydropower, solar, wind, biomass & waste, geothermal, wave, and tidal sources.



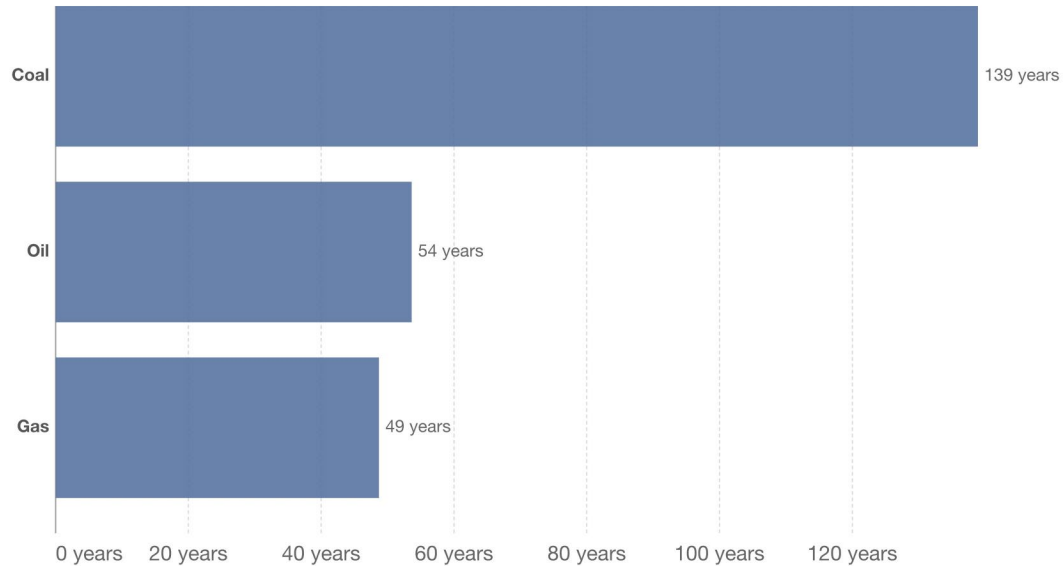
Source: Our World in Data based on BP Statistical Review of World Energy (2022); Ember; Our World in Data based on Ember's European Electricity Review (2022).
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Years of fossil fuel reserves left, 2020

Our World
in Data

Years of global coal, oil and natural gas left, reported as the reserves-to-product (R/P) ratio which measures the number of years of production left based on known reserves and present annual production levels. Note that these values can change with time based on the discovery of new reserves, and changes in annual production.

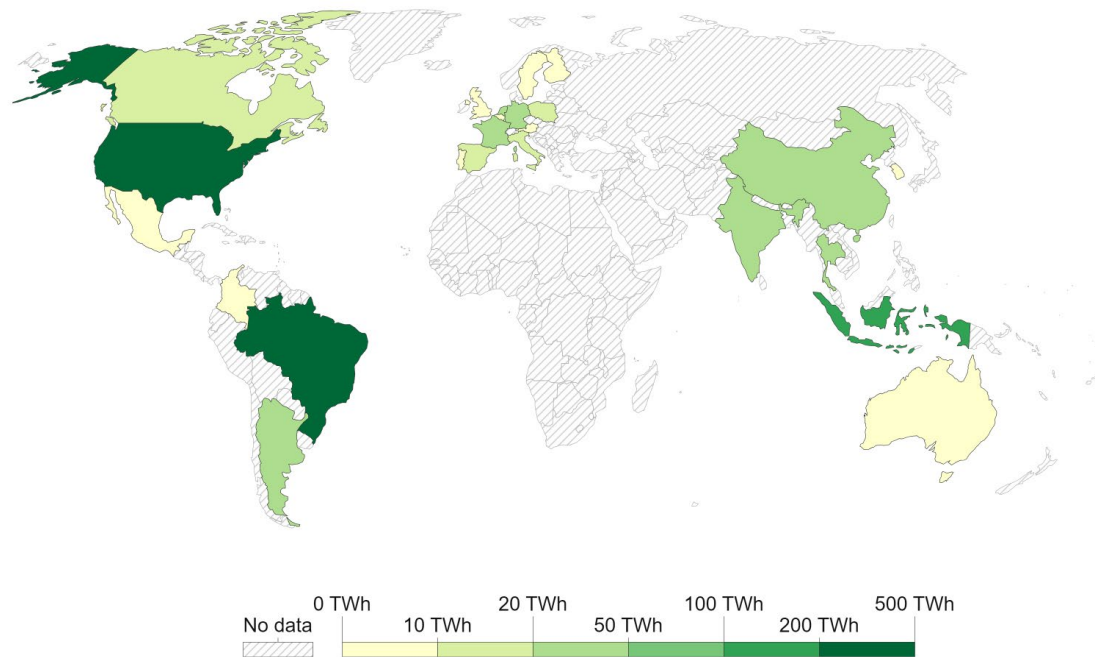


Source: BP Statistical Review of World Energy

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Biofuel energy production, 2022

Total biofuel production is measured in terawatt-hours (TWh) per year. Biofuel production includes both bioethanol and biodiesel.



Source: Energy Institute Statistical Review of World Energy (2023)

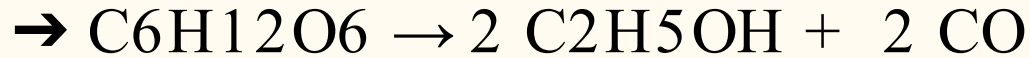
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Biofuel Explained

- From organic matter, such as plants, crops, algae, and organic waste
- Renewable and Sustainable



Example: Bioethanol



→ E10 Petrol is widely used



Nanotechnology in Biofuel

- Catalysts and Process Enhancements
- Enzyme Immobilization
- Nanostructured Materials for Fuel Storage
- Nanosensors for Quality Control
- Combustion Enhancements
- Engine Coatings and Lubricants
- Pollution Control

Experiment Materials

Different biofuels (e.g., biodiesel, bioethanol)

Various nanomaterials (e.g., metal nanoparticles, metal oxides)

Combustion chamber (compatible with the experimental setup)

Measurement instruments (e.g., emissions analyzer, calorimeter)

Test fuels without nanomaterials for comparison

Key Takeaways

- Biofuels are renewable, reduce carbon emissions, and enhance energy security.
- Nanotechnology improves biofuel production processes and combustion efficiency.
- Nanoparticles act as catalysts, enhance combustion, and reduce emissions.
- Nanostructured coatings and filters enhance engine efficiency and pollution control.

Questions

→ Any Qs?

